

Barbu Alexandru Mihai

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I'm a mechatronics and robotics engineer with a genuine passion for building things that actually work. Robotics drew me in from the start, but (jumping on the bandwagon) I am fascinated by AI and the quick growth it had over the last few years. I am curious about the next big step in robotics which would be about robots that actually perceive the world and how that would work. That's why my dissertation is about human-robot interaction and why I am seriously considering a PhD to dig deeper into this topic.

TECHNICAL SKILLS

Programming Skills:	Python, C/C++, JavaScript, MATLAB
AI & ML Skills:	YOLO, TensorFlow/TensorFlow Lite, PyTorch, Computer Vision, Model Quantization
Software Skills:	Fusion360, AutoCAD, SolidWorks, Inventor, Docker, Git, ROS2
Hardware Skills:	Raspberry Pi, Arduino, ESP32, Jetson Orin, Dynamixel Motors, Ip Cameras, IoT
Networking Skills:	MQTT, TCP/IP, UDP, Radio Communication, Cisco Packet Tracer, Subnetting
Soft Skills:	Technical Leadership, Team Mentorship, Problem Solving, Growth Ethic
Languages:	Romanian (Native), English (Fluent)

EDUCATION

2025 - 2027 (Expected)	Master's Degree in Robotics and Automation (Automatica si Calculatoare) at University Politehnica of Bucharest Customized curriculum drawing half the courses from the Artificial Intelligence Master program — covering Computer Vision, NLP, and ML theory — to complement the robotics core with the areas of AI I find most compelling.
2021 - 2025	Bachelor's Degree in Mechatronics and Robotics at University Politehnica of Bucharest Capstone Project: Delta Robot with Computer Vision for Object Sorting

EXPERIENCE

Telecommunications Engineer at MarcTel S.I.T. *November 2025 – Present*

- **Robotic Arm Integration:** Built and programmed a 6-DOF robotic arm using Dynamixel motors and ROS2 on a Jetson Orin NX. Took ownership of the control logic end-to-end — defined and stored key poses, then implemented smooth trajectory interpolation between them so the arm moves predictably without hitting the drone frame or any surrounding structure. The whole assembly sits on a drone platform, so collision-safe movement was a real constraint, not just a nice-to-have.
- **System Integration:** Integrated the arm and Jetson into a drone assembly alongside a Holybro Pixhawk 6 flight controller, establishing a stable UDP communication link between the two. Also integrated a radio station into the setup to allow remote wireless connection from a laptop to the full assembly.
- **Ground Control Application:** Designed and built from scratch a full PyQt5 desktop application for remote arm control. The app sends joystick commands over MQTT through the radio link, which the robot then listens to and executes. Also wired up an Arduino MKR with physical joysticks and buttons as a more ergonomic input device. One of the harder parts of the whole project — the arm control code had to handle a state machine with flying/landing/control modes, smooth multi-waypoint navigation, per-axis movement limits, gripper feedback over MQTT, and analog joystick deadzone handling, all

running concurrently in a ROS2 loop.

- **Adaptive Radio Power Control:** Wrote a background service that monitors RSSI from the radio link and automatically adjusts transmit power to maintain signal quality — runs transparently behind the GUI without any user interaction needed.

IoT Engineer at Vision Technology Development

June 2024 – October 2025

- **Smart Parking System:** Developed and deployed a C-based edge application for AXIS cameras using Docker, featuring a JavaScript interface for parking space management. Implemented MQTT communication for real-time status updates, with license plate recognition (LPR) on a secondary edge device and SQLite database integration for historical data.
- **Office Security:** Built a facial recognition system with anti-spoofing measures, including background comparison for edge detection and Time-of-Flight sensor integration for liveness detection.
- **Technical Mentorship:** Led a team of interns building a PTZ camera with fire detection capabilities — guided them through 3D design, dataset curation, model quantization, and motor control.
- **Fiber Optic Infrastructure:** Designed WAN/LAN networks using various networking equipment and technologies, with a working sample deployed in the lab for testing.
- **Computer Vision Research:** Developed a littering detection application using TensorFlow Lite with person tracking via DeepSort, with focus on model quantization and C implementation.

PROJECTS

Delta Robot with Computer Vision for Object Sorting

2024 – 2025

- Designed, 3D-printed, and assembled a fully functional delta robot capable of sorting small objects by type using an electromagnet.
- Trained a custom YOLOv8 model on self-collected images of nuts and washers for accurate object detection.
- Implemented a complete system on Raspberry Pi 5 — real-time detection, position calculation, inverse kinematics, and stepper motor control.
- Developed path planning and collision avoidance algorithms to optimize the sorting cycle.
- Ran simulations in MATLAB and Fusion360 to verify motor specs, trajectory performance, and structural load tolerance.

Plant Disease Detection using Deep Learning in MATLAB

Dec 2023 – May 2024

- Built a GUI that identifies diseases in tomatoes, potatoes, and bell peppers with diagnostic information and treatment recommendations.
- Trained a neural network on 9000 high-resolution plant images using Microsoft Azure for cloud compute.
- Integrated an ESP32-CAM stream into the interface, mounted on a custom mini-rover for field image capture.
- **Won 1st place** at “Zilele Educatiei Mecatronice 2024” in the Student Achievements – MATLAB Applications category.

Stroke Prediction System with Machine Learning

2024

- Processed and cleaned a dataset of 4000 stroke cases with mean imputation and SMOTE for class imbalance.
- Evaluated multiple ML algorithms; selected XGBoost based on AUC-ROC performance.
- Developed a clinical prediction GUI for inference on new patient data.

COURSES WITH CERTIFICATION

Oct 2025 - Dec 2025	CCNAv7: Enterprise Networking, Security and Automation
Dec 2024 - Jan 2025	Supervised Machine Learning: Regression and Classification
Dec 2024 - Jan 2025	Interfacing with the Arduino/Raspberry Pi
Oct 2024 - Jan 2025	Python101: Modules, OOP, Web-Scraping
Mar 2024 - Jun 2024	CCNAv7: Switching, Routing, and Wireless Essentials
Nov 2023 - Feb 2024	CCNAv7: Introduction to Networks
Oct 2023 - Jan 2024	Solid Edge Associate Level Certification

AWARDS

Dec 2025	1st Place — CodeQuest Hackathon, Artificial Intelligence Category Organized by Serviciul de Informații Externe. Built a real-time facial recognition pipeline capable of identifying individuals from a database at up to 12 meters, with or without masks. Fine-tuned YOLOv8n on a custom augmented dataset for masked/unmasked detection; benchmarked SAHI for improved recall; implemented a lightweight SVM classifier that reduced inference latency by ~90% compared to ArcFace and FaceNet512.
Apr–May 2024	1st Place — Zilele Educatiei Mecatronice 2024 Student Achievements in the Field of Mechatronics – MATLAB Applications. Presented the plant disease detection system deployed on a field rover.
Apr 2024	Excelence — A 46-a Sesiune de Comunicări Științifice Studentești (11–13 April 2024) Won the undergraduate scientific communications session with the plant disease detection GUI — approximately one month before ZEM 2024.